MIL-I-19210C(OS)
10 February 1976
Superseding
MIL-I-19210B(OS)
1 July 1968
and
Amendment 3
20 August 1971

MILITARY SPECIFICATION

IGNITER, TORPEDO, MARK 6 MOD 4

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of Torpedo Igniter Mk 6 Mod 4 (LD 160265), which provides the ignition in the combustion systems of steam and chemical type torpedoes. This igniter is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

* SPECIFICATIONS

Federal

TT-L-58	Lacquer; Spraying, Clear and Pigmented (General Interior Use)
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application
MIL-I-18997	Indicator, Pressure, Panel Mounted or Case Supported, General Specification

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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* MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with appropriate test methods

* DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

LD 160265	Torpedo Igniter Mark 6 Mod 4
LD 161803	Preserved (Packaging) Condition (Igniter Mark 6 Mod 4)
LD 161804	Shipping (Packing) Condition (Igniter Mark 6 Mod 4)
TG-10246	Torpedo Igniter Mark 6 Mod 4 Complete Assembly
TP-50056	Screw
TP-50700	Igniter Stock
TP-50712	Disc
TP-50104	Igniter Plug
749251	Screw
770276	Plug

PUBLICATIONS

* Naval Sea Systems Command (Code Ident 10001)

OD 9090	Methods for Loading, Assembling and Testing of Igniter Mark 6 Mod 4

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated,

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the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publication

Handbook H28 Screw-Thread Standards for Federal Services

(Applications for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The igniter is a combined pistol and double-fuze type device used to provide ignition for the combustion system of chemical and steam torpedoes.

3.2 General requirements.

* 3.2.1 Preproduction samples. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of the igniter is required and shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 for the purpose of determining, prior to production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract (see 4.3.1).

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the igniter shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 160265.

3.2.3 Finishes, coatings, construction, dimensions, and tolerances. Finishes, coatings, construction, dimensions, and tolerances for the igniter shall be strictly in accordance with the applicable drawings, specifications, and publications listed in LD 160265.

3.2.4 Moisture, fungus, and corrosion. The igniter shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications in LD 160265.

3.2.5 Marking. Marking of containers, assemblies, subassemblies, and parts shall be in accordance with this specification, the applicable drawings listed in LD 16025 and Drawings TG-10246.

3.2.5.1 Re-marking. Igniter assembly and shipping containers are reuseable and subject to return to loading activities. Prior to reloading igniters, marking such as "For Dynamometer Use Only" or other temporary marking shall be removed (see 5.3). Labels and other marking on shipping containers shall be removed and replaced with new markings as specified herein or in the contract (see 5.3).

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* 3.2.6 Loading. Methods of loading are prescribed in OD 9090.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

* 3.2.8 Conflicting requirements. In the event of conflict between the documents referenced herein and this specification, the requirements of this specification shall apply.

3.3 Definitions. For the purpose of this specification the following definitions apply:

3.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravity units (g). The frequency is given in Hertz (Hz).

3.3.2 Igniter axes. The axes of the igniter are:

(a) Longitudinal (axial)

(b) Perpendicular and transverse to (a)

(c) Mutually perpendicular to (a) and (b)

3.4 Performance requirements and product characteristics. The igniter shall meet the following performance requirements and product characteristics:

3.4.1 Leakage test requirements. Prior to installing and soldering Screw TP-50056 in place and assembling Plug 770276 in place, as specified on TG-10246, the igniter shall be subjected to leakage test procedures and methods prescribed in OD 9090 and shall meet the following requirements:

* 3.4.1.1 With igniter installed in test fixture as specified in OD 9090 and with 300 psia pneumatic pressure applied to outside surface of Disc TP-50712, Screw 749251, and Igniter Stock TP-50700, there shall be no leakage of air through the test port or at the end seal, as determined by the test procedure.

CAUTION: PRESSURE APPLICATION FOR TEST OF 3.4.1.2 SHALL NOT EXCEED 30 PSIA.

3.4.1.2 With 25 +5 -0 psia pneumatic pressure applied through the nipple of Igniter Plug TP-50104, there shall be no leakage of air through the test port of the 1-3/8 - 24 thread, as determined by the test procedures in OD 9090.

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3.4.1.3 With igniter installed in the test fixture as specified in OD 9090, and with 135 ± 5 psia hydrostatic pressure applied at the test port screw (TP-50056), there shall be no leakage of water by the test port screw.

3.4.2 Firing. Both chambers of the igniter shall fire on the first attempt when subjected to an operating air pressure of from 175 psi to 275 psi.

3.4.3 Duration. The igniter shall burn, with a steady flame, for a minimum of 25 seconds when fired in atmospheric pressure.

3.4.4 Length of flame. The minimum length of flame shall measure 4 inches from the end of igniter body to the tip of the flame.

3.5 Environmental requirements. The igniter shall meet all requirements of this specification during or after the environmental tests specified in 4.7.3.

3.5.1 Vibration. The igniter, packed and packaged in accordance with Section 5, shall be capable of withstanding the vibration testing specified in 4.7.3.1 and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.5.2 Shock. The igniter, packed and packaged in accordance with Section 5, shall be capable of withstanding the drop test specified in 4.7.3.2 and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.6 Cleanup. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. The igniter, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, coating, machine screw assemblage, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. Preproduction inspection (see 4.4).
2. Quality conformance inspection (see 4.5).
3. Periodic production inspection (see 4.6).

* 4.3 Sampling.

4.3.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of 15 igniters, manufactured in accordance with 3.2.1, shall be subjected to the preproduction inspection specified in 4.4 at a facility designated by the procuring activity (see 6.2). Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the igniter by the contractor, prior to approval of the preproduction sample, shall be at the contractor's risk.

* 4.3.2 Lot size. To ensure homogeneous lots, the inspection lot size shall be determined by the homogeneity of the various mixes or components which comprise the igniter. For example, if the batch size of one of the powder mixes is the limiting factor, that will establish lot size. Or if the quantity of other components such as diaphragms, firing pins, springs etc., is the limiting factor, that will determine the lot size. No inspection lot of igniters shall contain more than one batch of mixes or components.

* 4.3.3 Periodic production sampling. From each completed inspection lot that has satisfactorily passed the quality conformance inspection of 4.5, a periodic production sample shall be selected at random in accordance with the sampling plan of TABLE I. The sample igniters shall be subjected to the periodic production inspection of 4.6 at an activity designated by the procuring activity (see 6.2). Each igniter shall be identified by the lot number from which selected. The inspection lot will be accepted or rejected on the basis of the tests of 4.7 and the criteria of TABLE I. The acceptance test plan is based on Level II of MIL-STD-105, double sampling, and an acceptable quality level (AQL) of 1.0 percent.

* 4.4 Preproduction inspection. The preproduction sample igniters, after having satisfactorily passed the quality conformance inspection of 4.5, shall be subjected to the tests of 4.7. When specified in the contract or order (see 6.2), certified inspection data covering the results of the preproduction inspection shall be forwarded to the procuring activity. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial numbers of the igniter.

(d) Results of preproduction tests.

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TABLE I. Periodic production sampling plan

Inspection Lot Size	Type Sampling	Sample Size	Number of Defectives Accept	Reject
Up to 150*	Single*	13	0	1
151 to 280	Double	1st 32 2nd 32 Combined 64	0 1	2 2
281 to 500	Double	1st 32 2nd 32 Combined 64	0 1	2 2
501 to 1200	Double	1st 50 2nd 50 Combined 100	0 3	3 4

*Single sampling for lot size of 150 or less

* 4.5 Quality conformance inspection. Each igniter to be submitted on contract or order, including periodic production samples, shall be subjected to the examinations of 4.5.1 and 4.5.2. Manufacturer's certification that materials used are as specified in the applicable drawings shall be used in lieu of material tests.

* 4.5.1 Visual and dimensional examination. Each igniter shall be visually and dimensionally examined to verify conformance with the applicable drawings and the requirements for marking, cleanup and workmanship.

* 4.5.2 Preparation for delivery examination. Igniter packaging, packing and marking shall be examined to verify conformance to the requirements of Section 5.

* 4.6 Periodic production inspection. The periodic production sample igniters, selected in accordance with 4.3.3 and prepared for delivery in accordance with Section 5, shall be subjected to the tests of 4.7.

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* 4.6.1 Lot rejection report. When specified in the contract or order (see 6.2), when an inspection lot is rejected, a detailed report and test record of the inspection lot shall be submitted to the procuring activity and the lot held, pending further instruction from that activity.

4.7 Tests.

4.7.1 Test equipment. The following items of test equipment are required to perform the tests set forth in this specification:

(a) A test stand so constructed as to shield the igniter flame from drafts and to permit observation of the flame. The stand shall also be equipped with a suitable marker, fixed 4 inches from the firing end of the igniter.

(b) A source of compressed air capable of producing an operating pressure of 300 psia within 10 seconds.

(c) Valve, globe type, 1/2 inch.

(d) Gage, pressure, dial indication, circular face (MIL-I-18997), 0-300 psi accuracy of 5 percent.

(e) Stop watch, or similar device.

(f) Tubing, connecting, 1/8 inch I.D.

4.7.2 Test conditions. Unless otherwise specified, the igniter shall be subjected to tests under the following conditions:

(a) Temperature: Room ambient 65°F to 95°F (18°C to 35°C).

(b) Altitude: Normal ground.

(c) Vibration: None.

(d) Humidity: Room ambient to 95 percent relative maximum.

4.7.3 Environmental tests.

4.7.3.1 Vibration. The igniter, packed and packaged in accordance with Section 5, shall be subjected to the vibration testing of TABLE II in the three mutually perpendicular directions of the igniter axes defined in 3.3.2.

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TABLE II. Vibration testing schedule

Frequency Range	Amplitude	Minimum Test Duration Room Temp.	-20°F	+130°F
10-30 Hz	0.060 ±0.006 in.	60 min.	15 min.	15 min.
31-60 Hz	3 g ±0.3 g	60 min.	15 min.	15 min.
61-100 Hz	3 g ±0.3 g	60 min.	15 min.	15 min.

The frequency shall be continuously varied between each of the limits given, at a uniform rate over a 3 to 5 minute period, for the duration of the test.

As an alternate method, the igniter shall be vibrated at specific frequencies differing by not more than 5 Hz in each of the test axes and covering the entire frequency range and amplitudes of TABLE II. The total test time shall be not less than that of TABLE II, and the test times at the specific frequencies shall be equally divided.

4.7.3.2 Shock. The igniter, packed and packaged in accordance with Section 5 shall be subjected to drop testing as follows: The container shall be dropped cornerwise on a solid concrete, stone, or iron surface, falling freely through a vertical distance of 30 inches. Prior to each drop, the container shall be suspended with the center of gravity vertically above the striking corner. The container shall be dropped eight times, one on each of the eight corners, with the ends of the container being alternated for successive drops.

4.7.3.3 At the conclusion of the above environmental tests, the igniter shall be subjected to the functional tests of 4.7.4.

4.7.4 Functional tests.

4.7.4.1 Leakage test. Leakage tests shall be performed in accordance with the methods and procedures specified in OD 9090 for the purpose of determining compliance with the requirements of 3.4.1.1, 3.4.1.2, and 3.4.1.3 of this specification.

4.7.4.2 Assembled igniter test procedure. Place igniter in test stand of 4.7.1(a) and connect the stand, through the control valve and gage, to the air supply. Apply air gradually by means of control valve until igniter fires. The stop watch shall be started at the moment of firing and the following data shall be taken and recorded:

(a) Firing pressure in pounds per square inch. The igniter must meet the requirements of 3.4.2.

(b) Duration of burning. The flame must meet the requirements of 3.4.3.

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(c) Character of flame (e.g., steady, flickering, etc.). The flame must meet the requirements of 3.4.3.

(d) Length of flame. The length must be within the requirements specified in 3.4.4.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation - packaging. Preservation-Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A

* 5.1.1.1 Cleaning. Unless otherwise specified (see 6.2), the igniter shall be cleaned and degreased with Alcohol.

5.1.1.2 Drying. Not applicable.

* 5.1.1.3 Preservative application. Unless otherwise specified in the contract or order (see 6.2), coat the entire igniter, except the exposed threads on Igniter Stock TP-50700, with two coats of TT-L-58, Type I, lacquer. Coat the igniter stock threads with grade 1 corrosion preventive compound of MIL-G-16173.

5.1.1.4 Unit packaging. Unit packaging of the igniter shall be accordance with LD 161803.

5.1.1.5 Intermediate package. None.

5.1.1.6 Cushioning. None.

5.1.2 Level B. Not applicable to this specification.

5.1.3 Level C.

5.1.3.1 Cleaning, drying, and preservation. Cleaning, drying, and preservation shall be in accordance with 5.1.1.

5.1.3.2 Unit packaging. Unit packaging shall be in accordance with LD 161803.

5.1.3.3 Intermediate package. None.

5.1.3.4 Cushioning. None.

* 5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior containers. The igniter, preserved and packaged as specified in 5.1.1, shall be packed in accordance with LD 161804. In addition, the boxed igniters shall be further packed in a nailed wood box conforming to PPP-B-621, Class 2, Type 2 load, in accordance with MIL-STD-1186.

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5.2.2 Level B.

5.2.2.1 The igniter, preserved and packaged as specified in 5.1.1, shall be packed in accordance with LD 161804. In addition, the boxed igniters shall be further packed in a nailed wood box conforming to PPP-B-621, Class 1, Type 2 load.

5.2.3 Level C.

5.2.3.1 The igniter, preserved and packaged as specified in 5.1.3, shall be packed in accordance with LD 161804.

5.3 Marking.

5.3.1 Special marking. Unless otherwise specified, packages shall be identified with labels specified on the packaging drawings. The labels shall be marked with the required information indicated thereon. In addition, igniters accepted for dynamometer use only shall have the shipping container marked on all six surfaces with waterproof marking fluid as follows in 3/8 inch letters: "FOR DYNAMOMETER USE ONLY."

* 5.3.2 Normal marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and LD 161804.

6 NOTES

6.1 Intended use. The igniter is intended to be used in Torpedoes Mk 14 Mods 3A and 5 and Mk 16 Mod 8.

* 6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) Quantity of igniters required, including test samples.

(c) When a preproduction sample is not required (see 3.2.1)

(d) Any exception to this specification.

(e) Any exception to the cleaning and preservative requirements (see 5.1.1.1 and 5.1.1.3)

(f) Required level of preservation and packaging (see 5.1)

(g) Required level of packing (see 5.2)

(h) Assigned activity for preproduction and periodic production inspections (see 4.3.1 and 4.3.3)

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(i) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

(j) That the safety precaution requirements of the Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP 5, should be made applicable.

* 6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.6.1	Lot rejection report	-
4.4	Certified inspection data	-

DIDs (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

* 6.3 Storage life. The igniter has a storage (shelf) life of 3 years and, thereafter, will meet the performance requirements of this specification, when packaged and packed in accordance with level A of section 5 and stored under the following conditions:

a. Temperature: -65 to 160°F (-53 to 71°C)

b. Humidity: 95 percent relative, maximum

6.4 Superseding data. This specification supersedes those parts of NAVORD OS 2532 pertaining to manufacture and tests. This specification in conjunction with NAVORD OD 9090 completely supersedes, for procurement purposes, NAVORD OS 2532.

* 6.5 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no responsibility whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

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NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A

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